

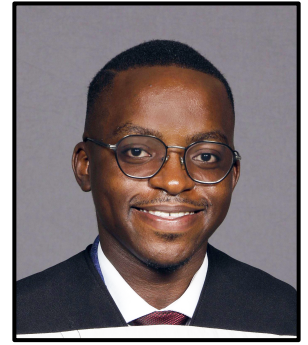
CV PAUL NAMALOMBA

🎓 **MSc(Eng) Structural, BSc(Hons.) Civil**

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🌐 <https://paulnamalomba.github.io> | [in](#) Paul Namalomba



RESUME SUMMARY

- Over **4.5 years of total technical experience** in Computational science and Digital infrastructure initiatives in Civil Engineering research and teaching at UCT, including end-to-end deployment of Computational Modelling software, usage of HPC clusters (Slurm, Rocky Linux). Further industry experience in building APIs in the .NET Framework (C#) and Django (Python). Broad experience in Data Engineering and Database Management with **PostgreSQL** + Application ORMs such as Django and NPGSQL, including innovating with large data updates with MPI + hyperthreading. My experience breaks down as:
 - 9 months experience** developing industry-grade APIs and Web Apps in **.NET Framework (C#)** and **TypeScript**, developing **Flutter** apps for Android and iOS, including a new ride-sharing app called **Ecoride Malawi**. Exploration of cloud-like (blob/object storage) for fast data manipulation and document retrieval. Implementing Self-Hosted Content Management Systems and Documentation Systems into web apps, using open-source platforms such as **Directus** and **Docmost**. Developed several company websites including Web Hosting and DNS management. My portfolio of websites designed include <https://computemore.com>, <https://ecoridemw.com>, <https://halfqr.com>. Obtained DevOps expertise in cloud-based deployment and orchestration of apps, APIs, backends and websites.
 - Nearly 2 (15 months) years' experience** architecting, maintaining, and optimizing the **SEAT UCT EBE enrollment platform** using Python Django with HTML, PostgreSQL, and LDAP for Security Management – We protected against SQL injections by using the Python Django layer for additional security. We innovated with data loads for up to 30 million data rows, by using MPI + hyperthreading, and batch-chunked memory management for effective server resource management at update time. Concept: [Batch Threading And Mpi For Large Db Ops - Paul Namalomba](#)
 - 2.5 years and counting** of project maintenance, and full-stack development of **SESKA finite-element computational suite** written in GNU C++, Fortran and GiD. First of its kind port from Linux to Windows whilst maintaining powerful Multiprocessing (Hyperthreading) compute methods.
- I also have more than **750 hours of active training** in Data Science, Data Engineering, High Performance Computing and Back-End Development.
- Led cross-functional teams of up to five teaching assistants in **lecturing** undergraduate students—aligning curriculum with emerging technologies and strategic learning objectives.
- Building my own python data processing packages utilizing **Computer Vision for Visual Data Analysis** with Google's Gemma 4 AI models, Random Forest outlier detection and other machine-learning innovations. (<https://libraries.io/pypi/datashadric>).

POINTS OF REFERENCE

- My entire portfolio is available online at my website: (<https://paulnamalomba.github.io>). Where I also publish guides and sample scripts.
- At my website I maintain guides on technical implementation and system file server management, as well as a history of all the technologies I have worked on **Web-Apps, Android/iOS Apps, Static Websites, Dashboards, WebSockets, Data Streaming Platforms** and **RESTful APIs**.
- My proofs of qualification are attached to this CV.

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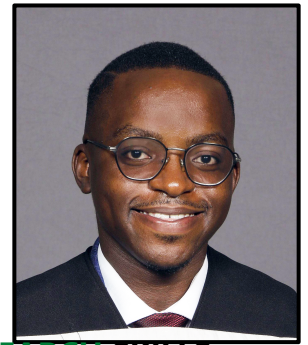
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CORE TECHNICAL SKILLS

- **Programming Languages:** C++, C#, React.js with Typescript, Python, SQL, Flutter, Swift
- **Web Scraping, Data Science and Statistical Data Analysis:** Python, Jupyter(*), Microsoft Power BI, Microsoft Tableau, Microsoft Excel, Python RESTful APIs
- **RDBMS, ETL Pipelines and Big Data (Cloud Computing):** Azure DB, PostgreSQL, MySQL, MongoDB, REDIS, Cloudflare R2 Blobs, Apache Kafka, Apache Airflow, Apache Hadoop DFS.
- **Programming IoT/Realtime Devices:** ESP-32 with MicroPython and C++, Micro-Sensor programming
- **REST APIs:** API programming with C# .NET and Python Django,
- **Realtime Data Streams and Communications:** SignalR WebSockets, Apache KAFKA and Firebase Cloud
- **DevOps and CI/CD:** GitHub, Bitbucket, Vercel, DockerHub and Cloudflare.
- **High-Performance Computing and Parallel Computing:** HPC, Slurm Workload Manager, OpenMPI, PETSc, METIS.
- **Data Security Policies:** LDAP Server Configuration, BCrypt, JWT Bearer, TLS/SSL configuration
- **Computational Modelling and Analysis:** SESKA, GiD, ABAQUS
- **Machine Learning and Regression Analysis:** Scikit-Learn, TensorFlow, statsmodels, datashadric
- **Advanced Shell Scripting, Automation, and Server Management:** Windows CMD Prompt, PowerShell 7+, Git BASH, MSYS2 Shells, Linux BASH.

PROFESSIONAL AND ACADEMIC EXPERIENCE

Professional: Chief Innovation Officer and Lead Backend Engineer

ComputeMore Africa Technologies Ltd (Tech Startup – [Company Website](#) | [Company GitHub](#)) | Blantyre, Malawi

10/2025 – Present | Experience: 9 months

***As Chief Innovation Officer I am tasked with consulting on Systems Design, proposing new projects and the double role of Lead Backend Engineer gives me first-class authority over Systems Architecture, Quality Control, Implementation and DevOps**

- **Achievements:**
 - Developed the Ecoride App API (a new ride-sharing app in Malawi looking to implement interesting Financial Technologies with Mobile Money Airtel and TNM). **Stack:** Flutter, Vite.js + Node.js (JavaScript), Redis, PostgreSQL, C# .NET (ASP & EF CORE), Google Maps API, PayChangu API, RabbitMQ and SignalR WebSocket (<https://ecoridemw.com>)
 - Developed [HalfQR](#), a smart, free QR-code generator with API capabilities for devs.
 - Leveraged **Cloudflare R2 Object Storage** for egress-free, low-cost cloud-like storage with custom archiving.
 - Built RestAPIs including Authentication, and Authorisation using **BCrypt** and **JWT** 256-bit tokens for security hardening. Clean architecture following the Controller -> DTOs -> Services -> Entities philosophy.
 - **DevOps** managed with MS Azure for hosting backends, Physical Server configuration for backends and frontends with Nginx, and Cloudflare, Vercel for websites and low-load frontends.
 - Our tech partners included Namecheap, Cloudflare for web hosting and blob/cloud storage (R2); PayChangu (Fintech provider), Ecoride Malawi (Client) and ICTAM.

OTHER TECHNICAL+RESEARCH SKILLS

- **Scientific Research Design, Reporting and Methodology**
- **The Engineering Design and Software Design Life Cycl**
- **AGILE Methodologies for Software Engineering Workflow**
- **A Multi-Disciplinary Depth of Knowledge (Computing, Data Science and Statistics, Engineering and Research Management)**
- **Graphics Design for Posters:** Adobe Photoshop, Canva.
- **Other Advanced ICT Skills:** Microsoft Office, LibreOffice, Microsoft OneNote.
- **Operating Systems:** Linux Ubuntu 20.04+, Windows 10+, WSL; Linux Ubuntu Server 20.04+,

RESEARCH INTERESTS

- **Computational Mechanics and Applied Mathematics**
- **Shelf-Ice Mechanics and Rheology**
- **Non-linear Large-Deformation Elasticity, Visco-Plasticity, and Elasto-Plasticity**
- **Quasi-Brittle and Ductile Failure Modes**
- **Computational Algorithm Optimization**
- **Data Analysis and Digital Infrastructures**
- **Google Maps APIs and REST APIs**
- **Emerging Financial Technologies**
- **High-Performance and Parallel Computing**

- (References: [Mr. Thanthwe Kalonga](#))

Professional: SEAT* Backend Application Developer (Hybrid) – Experience: 1.3 Years

EBE Faculty, University of Cape Town ([Faculty Website](#)) | Cape Town, South Africa

12/2025 – 02/2026 | 11/2024 – 07/2025 | 12/2021 – 01/2022

***SEAT is an in-house enrolment platform developed at the Faculty of Engineering & the Built Environment**

Front-end: Python Django with HTML Views. **Database Architecture:** PostgreSQL (psycopg2) - **Remote Server-Hosted.**

• Tasks and Achievements:

- **DevOps** managed with Git CLI, Linux BASH Shell (Ubuntu 24.04 Server) and Django CLI/Shell tools.
- Code review, debugging, re-factoring and module linking (Python Django, JavaScript, HTML)
- Data Engineering – managing database migrations, database updates, database access control and database resource management (PostgreSQL, pgAdmin, Django ORM)
- Developed an AI-based timetable generator
- Improved server availability to 100% with 0 server errors in a month
- Maintained client (end-user) verification and authorization protocols with LDAP
- Some UX design also was done, implementing new components to make it easier for students to see their outstanding enrolments
- Email automation with SMTP and Django's shell-based direct code-execution
- (References: [Prof. Sebastian Skatulla](#), [Ms. Celeste Wilson](#))

Academic: SESKA* Computational Engineer as part of Doctor of Philosophy: Civil Engineering

University of Cape Town | Cape Town, South Africa

09/2023 – Present | Experience: 2.5 Years

***SESKA is an in-house Finite Element Analysis software package maintained in the Department of Civil Engineering**

Back-End: C++, and Fortran. **HPC and Parallel Computing:** PETSc, OpenMPI, UCT HPC. OS: Linux, Windows

• Tasks and Achievements:

- **DevOps** managed with Git CLI, SmartGit GUI and BitBucket Web.
- Deployed SESKA onto UCT HPC, the EBE Computational Mechanics Labs, and Chemical Engineering Computer Labs with a Full-Cycle development of windows Port of SESKA with UCRT and MVSC++, that I led and completed solely.
- Implemented of working **elasto-plasticity**, **visco-plasticity**, **material damage**, **cyclic loading**, and **variable elasticity** models.
- Innovations include computational improvements using inline ternaries instead of nested conditionals, pre-incrementation for loops, and memory clean-ups for temporary quantities – preferring in-place computing.
- (References: [Prof. Sebastian Skatulla](#), [Mr. Sibonelo Twala](#), [Mr. Theo Armstrong](#), [Dr. Maxime Nutte](#))

Academic: Assistant Lecturer in Engineering Mechanics – Experience: 2 Years

University of Cape Town | Cape Town, South Africa

02/2025 – 07/2025 | 06/2024 – 12/2024 | 06/2023 – 12/2023 | 06/2022 – 12/2022

• Tasks and Achievements:

- 85%+ Course Pass Rate for CIV2011F (2025), 93%+ Course Pass Rate for CIV1007S (2022, 2023, 2024)
- Crafted the course outlines, course content and exams from Scratch.
- Led and directed a team of five Teaching Assistants, managed budgets and schedules
- Enhanced technical and practical instruction of students using ***SESKA**
- (References: [Prof. Sebastian Skatulla](#), [Ms. Ayesha Dalwai](#))

EDUCATION HISTORY

Ongoing Doctor of Philosophy: Civil Engineering

University of Cape Town | Cape Town, South Africa | 09/2023 – Present | Expected Completion 09/2026

- **Topic of study** – A Nonaffine Damage Theory for Multisolid Composites
- **Research Keywords:** Composite Materials, Viscoelasticity, Computational Mechanics, Applied Mechanics
- **Skills Developed:** Research Management, Research Design and Reporting, HPC, Microsoft Visual Studio C++, GNU C++, GNU Fortran, Mingw32/64 Cross-Compatibility Tools, Python for Data Science, Git, MSYS2 Shells and Linux BASH.

Master of Science in Engineering: Structural Engineering & Materials

University of Cape Town | Cape Town, South Africa | March 2020 – December 2021

- **Distinction in Thesis** – Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.
- **Research Keywords:** Material Damage, Computational Mechanics, Applied Mechanics

- **Skills Developed:** Research Management, Research Design and Reporting, HPC, GNU C+, Python for Data Analysis, Git and Linux BASH.

Bachelor of Science (Honours): Civil Engineering

University of Cape Town | Cape Town, South Africa | January 2016 – December 2019

- **Distinction in Dissertation** – *Design of A Climate Chamber for Uni-axial Compressive Strength Testing of Laboratory Grown Sea Ice.*
- **Dean's Merit List:** 2016, 2017.

LICENSES & CERTIFICATES

Ongoing Professional Certificate: IBM Data Engineering

Coursera | 07/2025 – Present | Expected Completion End 07/2026 | 8/16 Certificates Completed (150+ hours)

- IBM – Introduction to Data Engineering
- IBM – Python for Data Science, AI & Development
- IBM – Python Project for Data Engineering
- IBM – Introduction to Relational Databases (RDBMS)
- IBM – Databases and SQL for Data Science with Python
- IBM – Hands-on Introduction to Linux Commands and Shell Scripting
- IBM – Relational Database Administration (DBA)
- IBM – Introduction to Big Data with Spark and Hadoop

Ongoing Professional Certificate: Microsoft Back-End Developer

Coursera | 09/2025 – Present | Expected Completion End 09/2026 | 1/8 Certificates Completed (30+ hours)

- Microsoft – Foundations of Coding Back-End

Professional Certificate: Google Advanced Data Analytics

Coursera | 12/2024 – 05/2025 | 300+ hours of training

- Google - Foundations of Data Science
- Google - Regression Analysis: Simplify Complex Data Relationships
- Google - The Power of Statistics
- Google - Go Beyond the Numbers: Translate Data into Insights
- Google - Get Started with Python
- Google - The Nuts and Bolts of Machine Learning

Developer Certificate: Data Analysis with Python

Coursera | 04/2025 – 05/2025 | 200+ hours of training

- Data Structures: Pandas DataFrames, Numpy Arrays
- Data Cleaning and Visualization: Matplotlib, Seaborn
- Data Scraping and Data Extraction: .csv, HTML, BigQuery

ORGANIZATIONS AND TECH SUBSCRIPTIONS

- **GitHub Pro Contributor** (Open-Source and Closed-Source Private/Enterprise Contributions) – 1500+ contributions (+500,000 lines of code) since June 2025 – <https://github.com/paulnamalomba>
- **PyPI Contributor and Original Innovator** (kabwenzenamalomba@gmail.com – diversecellar)
- **AfricasTalking API Subscription** (kabwenzenamalomba@gmail.com), **Airtel Africa API Developer Platform** (kabwenzenamalomba@gmail.com)
- **Upstash Context7 API Subscription** (kabwenzenamalomba@gmail.com)
- **Android Studio Developer Platform** (kabwenzenamalomba@gmail.com); **Google Cloud Developer Platform** (kabwenzenamalomba@gmail.com); **Google AI Pro Subscription** (kabwenzenamalomba@gmail.com)
- **South African Statistical Association**, July 2025 – Present [ID 2748543]
- **Statistics Without Borders**, January 2025 – Present [ID 16089]
- **South African Institution of Civil Engineering**, 2021 – Present [ID 202100183]

CONFERENCES & PUBLICATIONS

- **MSc Thesis 2022** [available: <http://hdl.handle.net/11427/37632>]: Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.
- **Africomp7 conference, February 2026** – scheduled to present two abstracts:
 - **A Lie Group Approach to Non-Affine Interaction and Damage in Multi-Solid Composites** with **Prof. Sebastian Skatulla**, **Prof. Michael Kaliske**, **Prof. Carlo Sansour**

- Logarithmic Strain-Based Glen's Flow Law for Glaciological Applications with [Prof. Sebastian Skatulla](#), [Prof. Michael Kaliske](#), [Prof. Carlo Sansour](#), [Dr. Maxime Nutte](#)
- ICMM7 Conference, November 2022.
- SACAM2020 Conference, November 2021.

COMMUNITY INVOLVEMENT & LEADERSHIP

- CRC Cape Town (Malawi): Service in Altar Team, July 2025 – Present.
- Redeemed For a Purpose Ministries (Malawi): Service in Financial Coordination, April 2025 – Present.
- International Students Forum (UCT): March 2025 – Present.
- Statistics Without Borders Volunteer, January 2025 – Present.
- Mentorship and Supervision of Honour's Researchers (UCT), June 2021 – Present.
- Volunteer Work for IAPO Soccer Tournament (UCT), October 2022.
- Flood and Disaster Relief Efforts (Mangochi, Malawi), January 2022.
- Child Education Advocacy (Mangochi, Malawi), January 2017 & January 2018.

REFEREES

Prof. Sebastian Skatulla^[1,2] – PhD Research Supervisor, MSc Research Supervisor, SEAT Lead Developer

[1] Professor of Structural Engineering & Materials | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

[2] Director | Computational Continuum Mechanics Research Group, University of Cape Town

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Mr. Theo Armstrong^[1,2] – Laboratory Administrator (UCT ICTS)

Laboratory Administrator | Student Computing Services, University of Cape Town

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Mr. Sibonelo Twala^[1,2] – Senior Systems Engineer (UCT ICTS)

Senior Systems Engineer | Workplace Services, University of Cape Town

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Ms. Celeste Wilson – Deputy Faculty Manager (UCT EBE)

Deputy Faculty Manager | Faculty of Engineering & the Built Environment, University of Cape Town

📍 Level 6 Admin, New Engineering Building, Upper Campus, Cape Town

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Ms. Ayesha Dalwai – Departmental Manager (UCT EBE – Civil Engineering)

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Dr. Keith MacHutcheon – MSc Research Supervisor

Honorary Research Associate | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

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Dr. Maxime Nutte – Research Collaborator

Postdoctoral Fellow | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

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Mr. Thanthwe Kalonga^{1,2]} – Managing Director of ComputeMore Africa Technologies Ltd

[1] Executive Managing Assistant and Actuarial Manager | NICO Life (Sanlam-Allianz)

[2] Managing Director | ComputeMore Africa Technologies Ltd

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